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PROJECT TASK - 4

DOCUMENTATION

REPORT

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7-Segment Display Driver



Abstract

This project implements a 7-Segment Display Driver using Verilog HDL. The circuit converts 4-bit binary input values into corresponding segment control signals required to display decimal digits from 0 to 9 on a 7-segment display.

The design demonstrates combinational logic implementation, binary-to-display conversion, and digital display interfacing. Simulation and waveform verification were performed using EDA Playground and EPWave.



Objective

The main objectives of this project are:

- To design a 7-Segment Display Driver using Verilog HDL
- To convert binary inputs into segment display outputs
- To understand combinational logic circuits
- To perform simulation and waveform verification
- To display decimal numbers using digital display systems



Tools Used

Tool	Purpose
Verilog HDL	Hardware Discription Language
EDA Playground	Online Simulation Platform
Icarus Verilog	Verilog Simulator
EPWave	Waveform Viewer



Theory

7-Segment Display

A 7-segment display is an electronic display device used to display decimal digits. It consists of seven LEDs arranged in the form of segments.

Segment Representation :

a
f b
g
e c
d

Each segment is controlled individually to display different numbers.

Binary to Segment Conversion Segment Driver Module

The module:
The driver circuit accepts:

4-bit binary input

and generates:

7-bit segment output

which controls the display segments.

Accepts binary input

Produces segment control output

Uses combinational logic

Implements case-based segment mapping

Example :

Binary Input	Displayed Number
0000	0
0001	1
0010	2
0011	3

Working Principle

- A 4-bit binary number is applied as input.
- The Verilog case statement checks the input value.
- Corresponding segment outputs are generated.
- The required segments turn ON to display the decimal digit.
- Waveform simulation verifies correct segment activation.

Verilog Module Used

Seven Segment Driver Module

The module:

- Accepts binary input
- Produces segment control output
- Uses combinational logic
- Implements case-based segment mapping

Flies Included

-  Main Verilog design code
-  Testbench code
-  Output screenshots
-  EPWave waveform screenshots

Simulation Procedure

1. Open EDA Playground
2. Select Verilog language
3. Paste design code into design.sv
4. Paste testbench into testbench.sv
5. Click Run
6. Open EPWave
7. Add signals:
 - num
 - seg



Output Waveform Analysis

The waveform output shows the relationship between:

- Input binary values
- Segment output patterns

As the input value changes from:

0000 → 1001

the segment outputs change accordingly to display decimal digits from:

0 → 9

The waveform verifies correct operation of the display driver.

TestBench Code, Main Code

EDA playground

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Doulos does not endorse training material from other suppliers on EDA Playground.

Languages & Libraries

Testbench + Design

SystemVerilog/Verilog

UVM / OVM

None

Other Libraries

None

OVL

SVUnit

Enable TL-Verilog

Enable Easier UVM

Enable VUnit

Tools & Simulators

Icarus Verilog 12.0

Compile Options

-Wall

Run Options

Run Options

Use run.bash shell script

Open EPWave after run

Show output text file

Show output SVG file

Download files after run

Examples

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testbench.sv

design.sv

```
1 // Testbench for 7-Segment Display
2 module testbench;
3
4 // Input
5 reg [3:0] num;
6
7 // Output
8 wire [6:0] seg;
9
10 // Instantiate module
11 seven_segment uut(
12
13     .num(num),
14
15     .seg(seg)
16 );
17
18 initial begin
19
20     // Generate waveform
21     $dumpfile("dump.vcd");
22     $dumpvars(0,testbench);
23
24     // Monitor outputs
25     $monitor(
26         "Number=%b Segment=%b",
27         num,seg
28     );
29 end
```

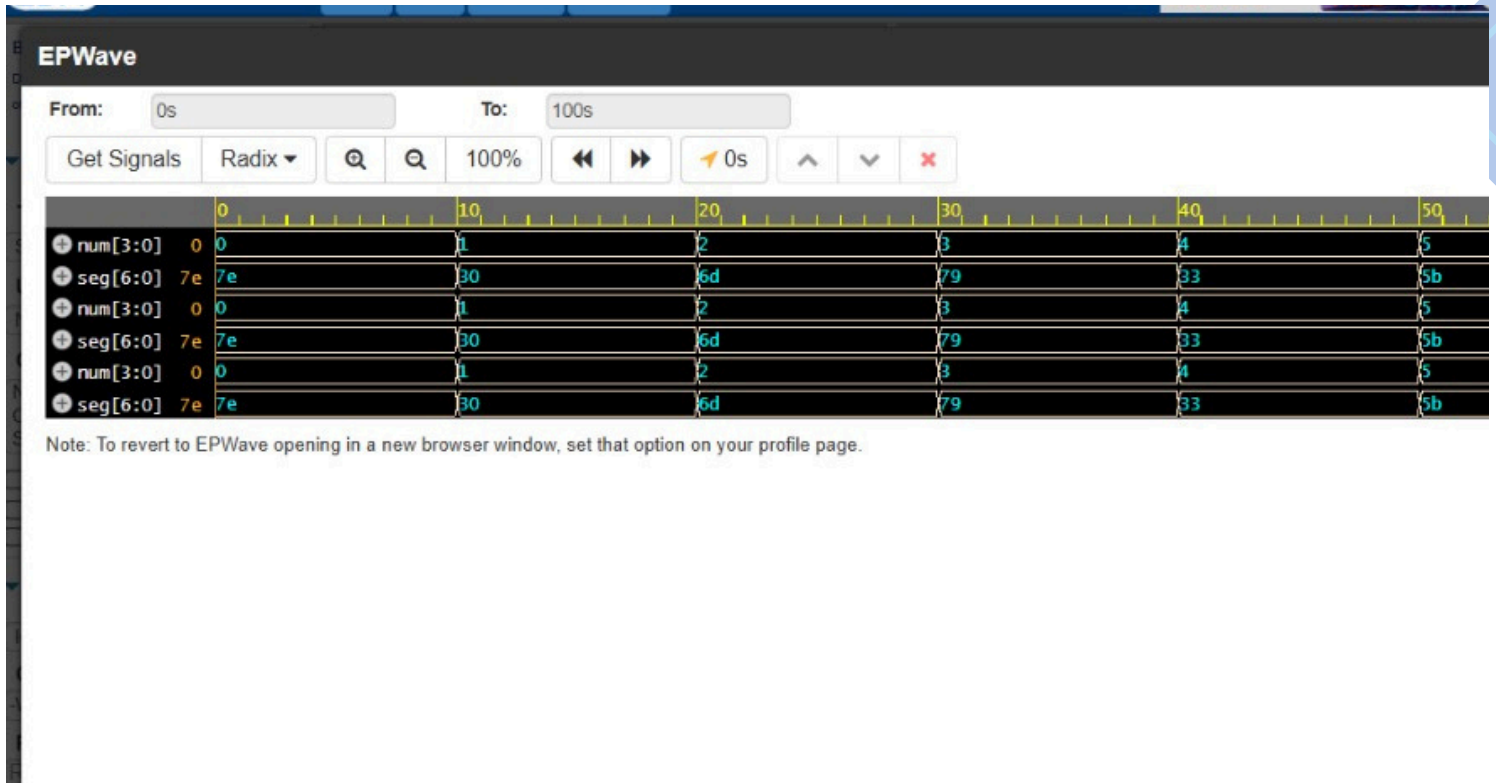
```
1 // 7-Segment Display Driver
2 module seven_segment(
3
4     input [3:0] num,
5
6     output reg [6:0] seg
7 );
8
9 // Segment control
10 always @(*)
11 begin
12     case(num)
13
14         // Display 0
15         4'b0000:
16             seg = 7'b1111110;
17
18         // Display 1
19         4'b0001:
20             seg = 7'b0110000;
21
22         // Display 2
23         4'b0010:
24             seg = 7'b1101101;
25
26         // Display 3
27         4'b0011:
28             seg = 7'b0110111;
29
30         // Display 4
31         4'b0100:
32             seg = 7'b1011011;
33
34         // Display 5
35         4'b0101:
36             seg = 7'b1011111;
37
38         // Display 6
39         4'b0110:
40             seg = 7'b1111011;
41
42         // Display 7
43         4'b0111:
44             seg = 7'b1110000;
45
46         // Display 8
47         4'b1000:
48             seg = 7'b1111111;
49
50         // Display 9
51         4'b1001:
52             seg = 7'b1111011;
53
54         default:
55             seg = 7'b0000000;
56     endcase
57 end
```

Log

Share

Number=0011 Segment=1111001
Number=0100 Segment=0110011
Number=0101 Segment=1011011
Number=0110 Segment=1011111
Number=0111 Segment=1110000
Number=1000 Segment=1111111
Number=1001 Segment=1111011
testbench.sv:72: \$finish called at 100 (1s)
Finding VCD file...
./dump.vcd
[2026-06-04 09:16:07 UTC] Opening EPWave...
Done

Logic Gates Design-EV Waveforms



References :

- Verilog HDL Documentation
- Digital Electronics Fundamentals
- EDA Playground Simulator Documentation

Conclusion :

The 7-Segment Display Driver was successfully designed and simulated using Verilog HDL. The waveform outputs verified proper conversion of binary input values into segment control signals. The project helped in understanding combinational logic design and digital display interfacing techniques.